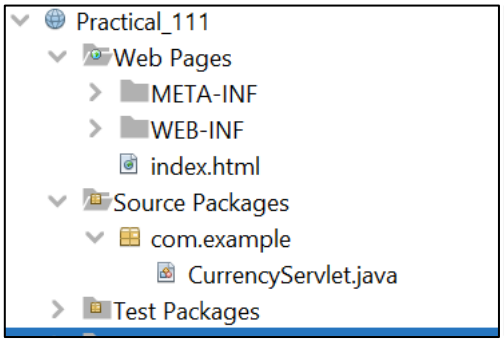
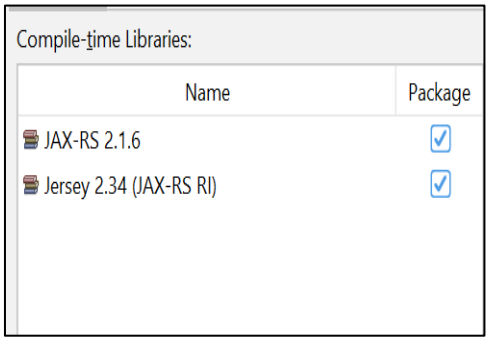
PRACTICAL 11

Simple RESTful Service Develop a simple JAX-RS service that provides currency conversion.





CODE:

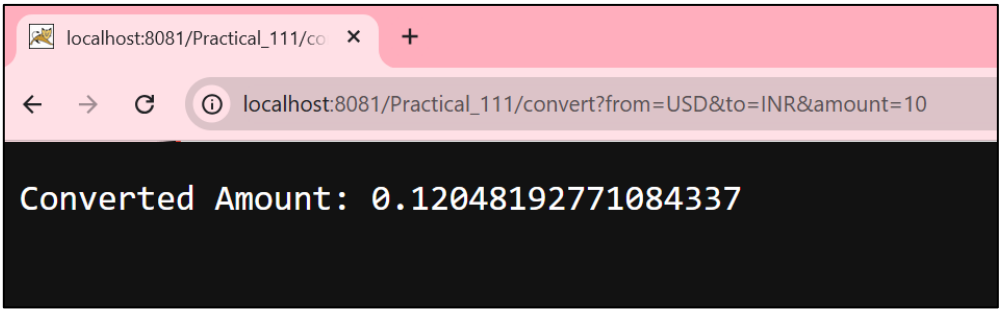
CurrencyServlet.java

```
//Simple RESTful Service
//Develop a simple JAX-RS service that provides currency conversion.

package com.example;

import java.io.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.*;
import javax.servlet.http.*;

@WebServlet("/convert")
public class CurrencyServlet extends HttpServlet {
```

```
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        String from = request.getParameter("from");
        String to = request.getParameter("to");
        double amount = Double.parseDouble(request.getParameter("amount"));

        double result = convert(from, to, amount);

        response.setContentType("text/plain");
        PrintWriter out = response.getWriter();
        out.println("Converted Amount: " + result);
    }
    private double convert(String from, String to, double amount) {
        double usd = 83, inr = 1, eur = 90;

        double fromRate = from.equalsIgnoreCase("USD") ? usd :
                        from.equalsIgnoreCase("EUR") ? eur : inr;

        double toRate = to.equalsIgnoreCase("USD") ? usd :
                      to.equalsIgnoreCase("EUR") ? eur : inr;

        return (amount / fromRate) * toRate;
    }
}
```

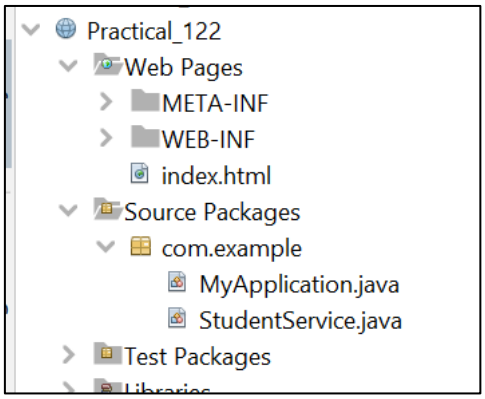
OUTPUT:

http://localhost:8080/Practical_111/convert?from=USD&to=INR&amount=10



PRACTICAL 12

Data Driven RESTful Service Develop a RESTful service to accept student registration data sent from a web form.



CODE:

Index.html

```
<!DOCTYPE html>
<html>
```

```
<head>
   <title>Student Registration</title>
</head>
<body>

<h2>Student Registration Form</h2>

<form action="http://localhost:8081/Practical_122/webapi/student/register"
method="post">

   Name: <input type="text" name="name"><br><br>
   Email: <input type="email" name="email"><br><br>
   Course: <input type="text" name="course"><br><br>

   <input type="submit" value="Register">

</form>

</body>
</html>
```

StudentService.java

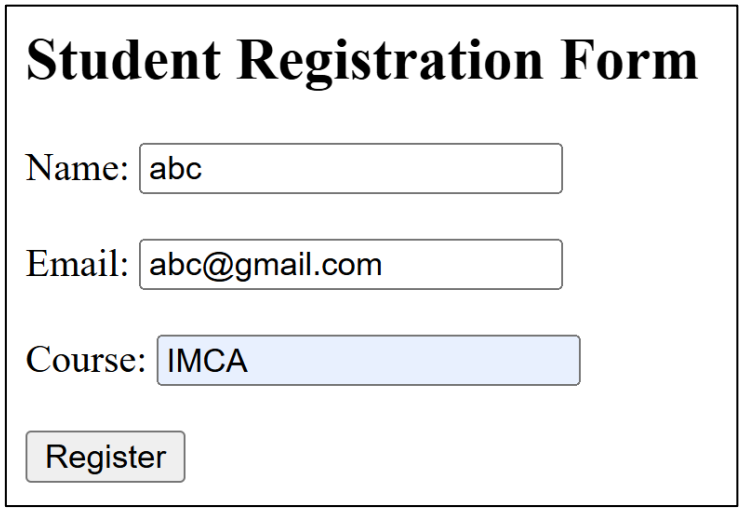
```
package com.example;
import javax.ws.rs.POST;
import javax.ws.rs.Path;
import javax.ws.rs.FormParam;
import javax.ws.rs.Produces;
```

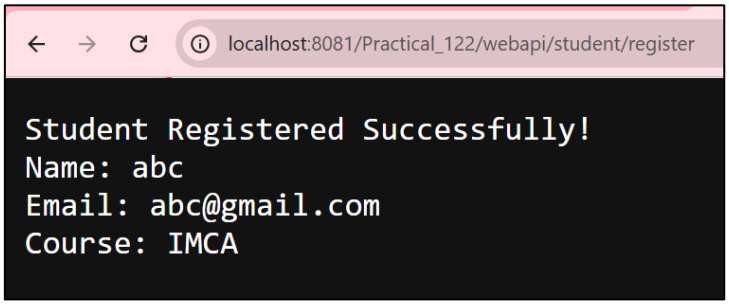
```
import javax.ws.rs.core.MediaType;

@Path("/student")
public class StudentService {

    @POST
    @Path("/register")
    @Produces(MediaType.TEXT_PLAIN)
    public String registerStudent(
            @FormParam("name") String name,
            @FormParam("email") String email,
            @FormParam("course") String course) {

        return "Student Registered Successfully!\n"
                + "Name: " + name
                + "\nEmail: " + email
                + "\nCourse: " + course;
    }
}
```

MyApplication.java

```
import javax.ws.rs.ApplicationPath;
import javax.ws.rs.core.Application;

@ApplicationPath("/webapi")
public class MyApplication extends Application {
}
```

OUTPUT:


Student Registration Form
Name: abc
Email: abc@gmail.com
Course: IMCA
Register


localhost:8081/Practical_122/webapi/student/register
Student Registered Successfully!
Name: abc
Email: abc@gmail.com
Course: IMCA